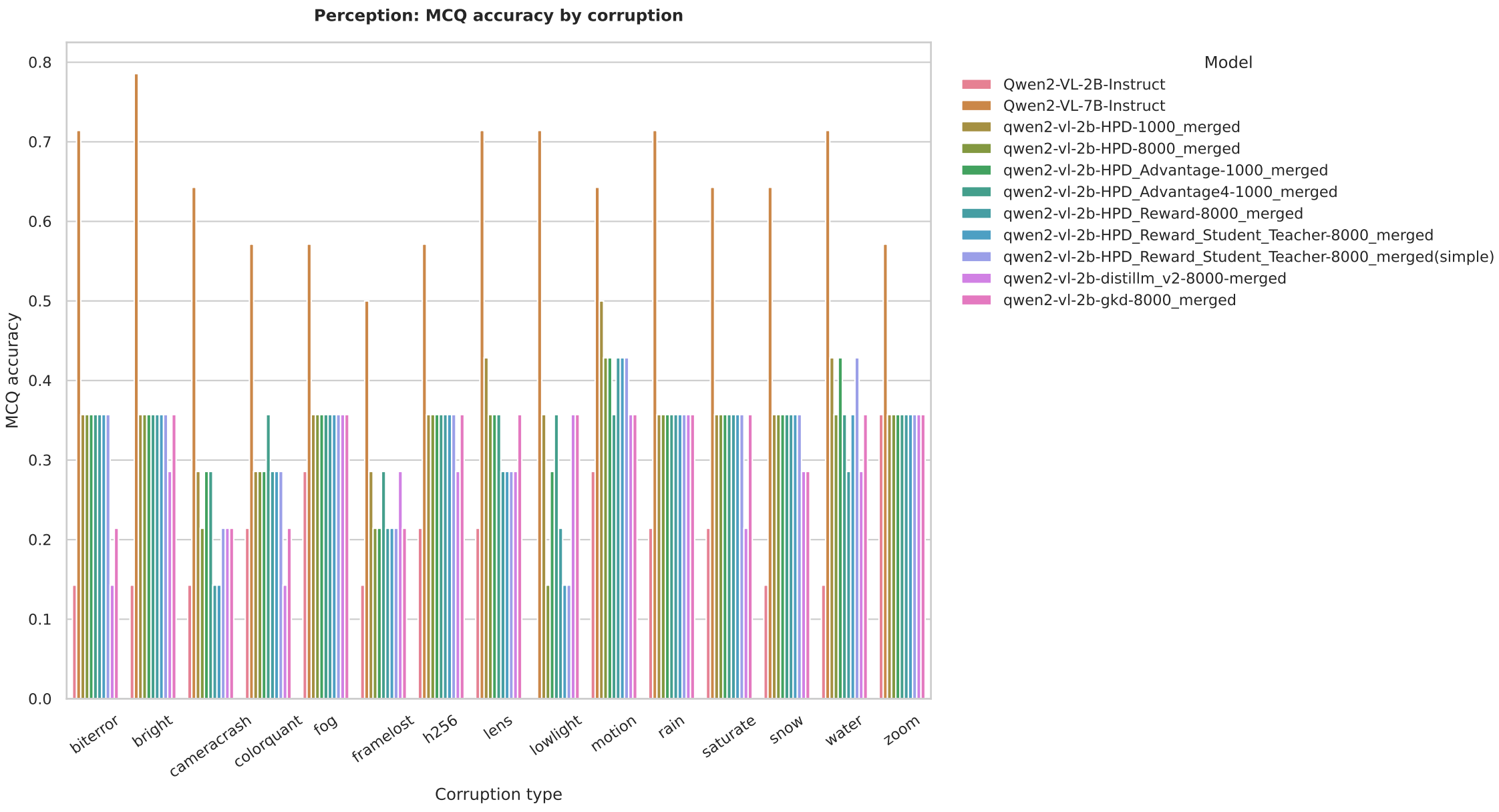
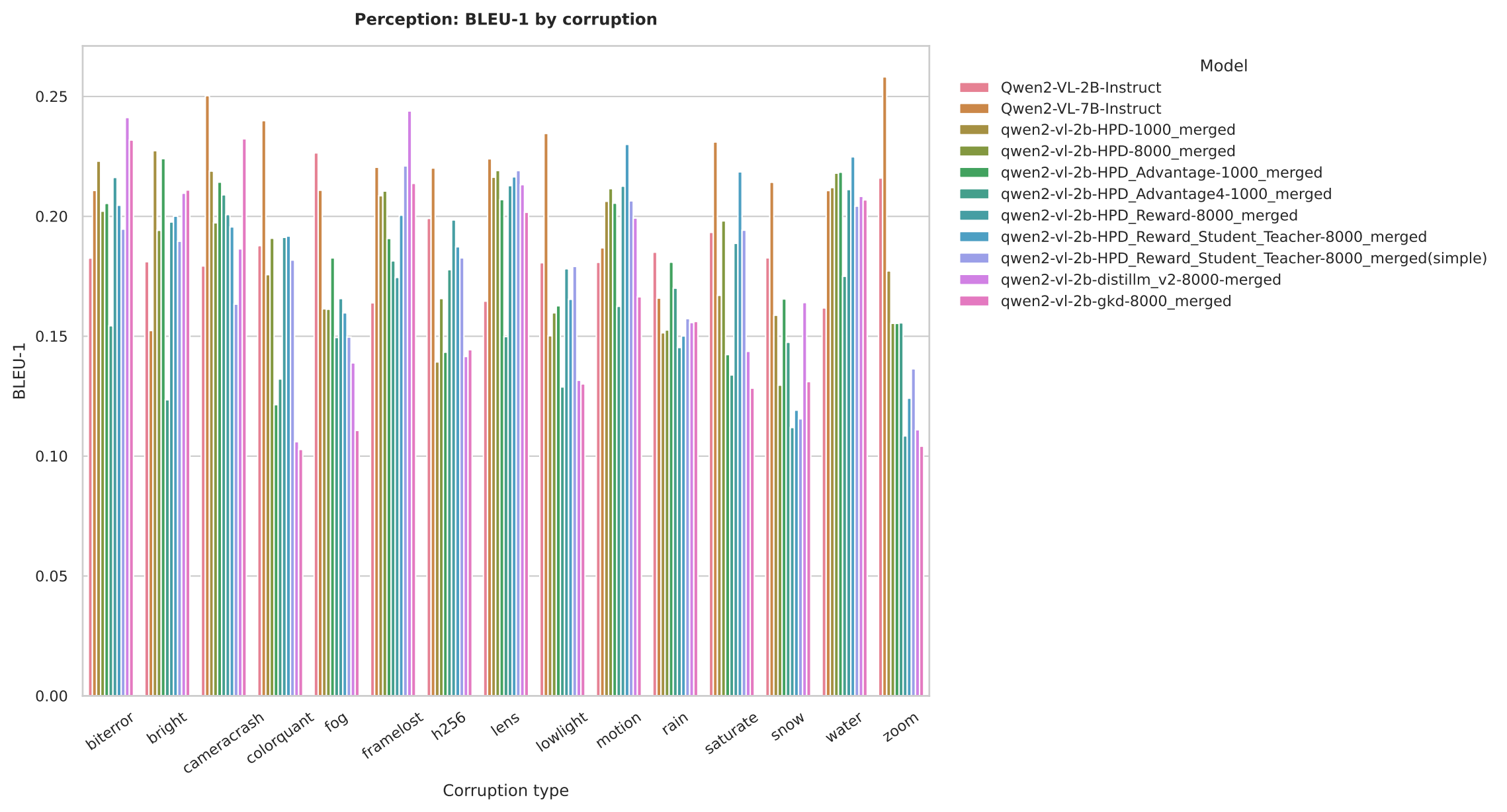
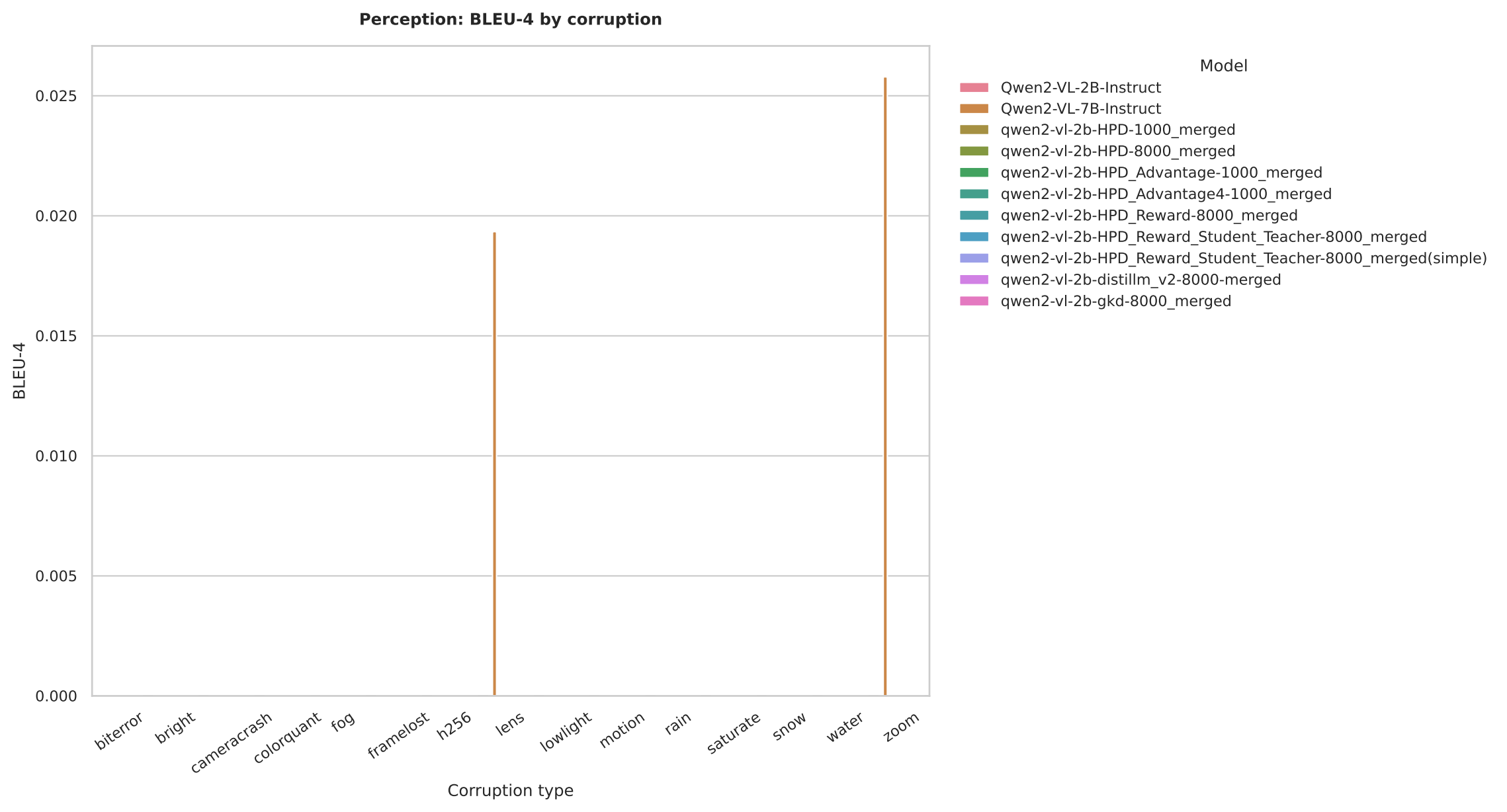
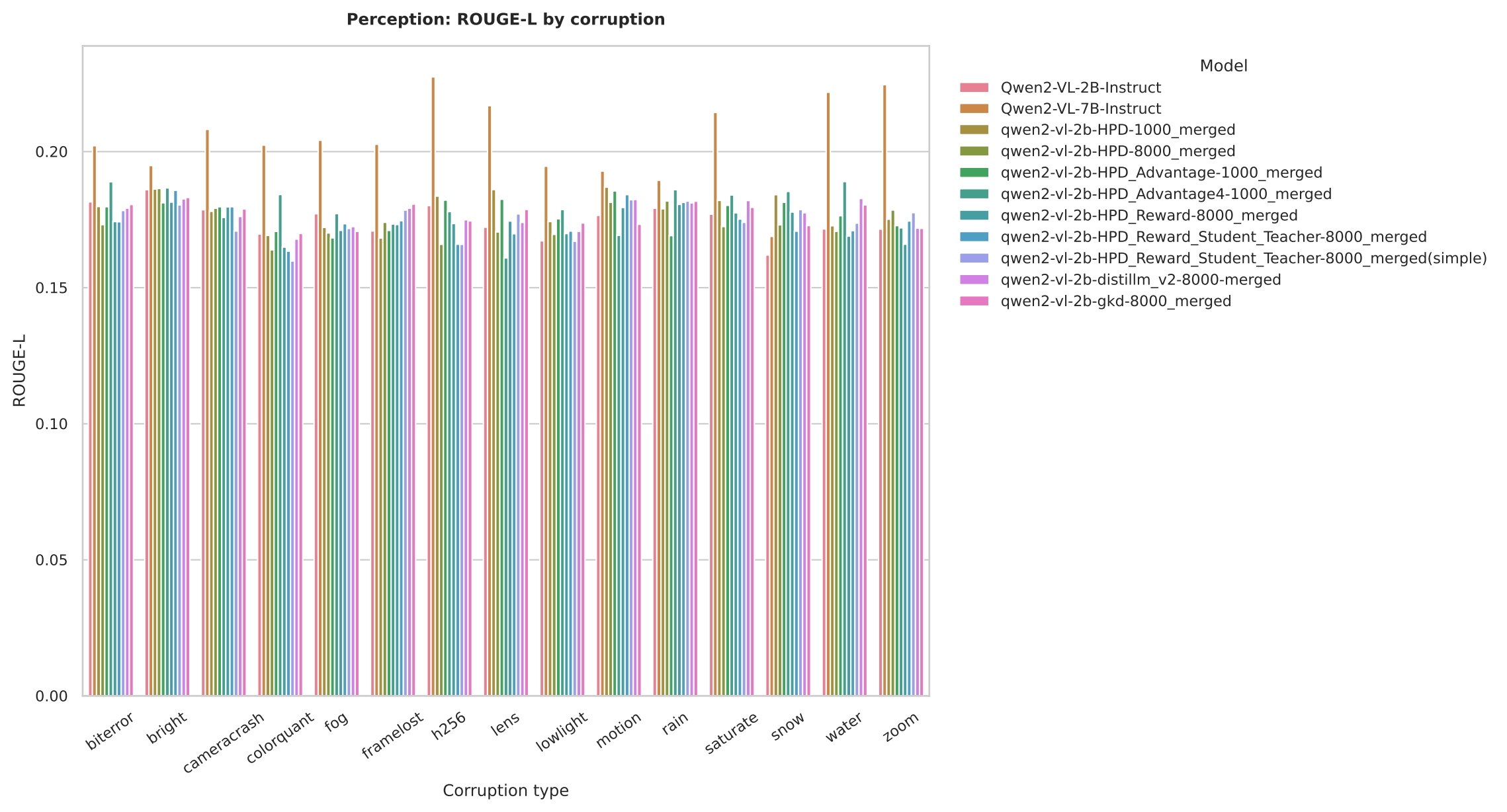
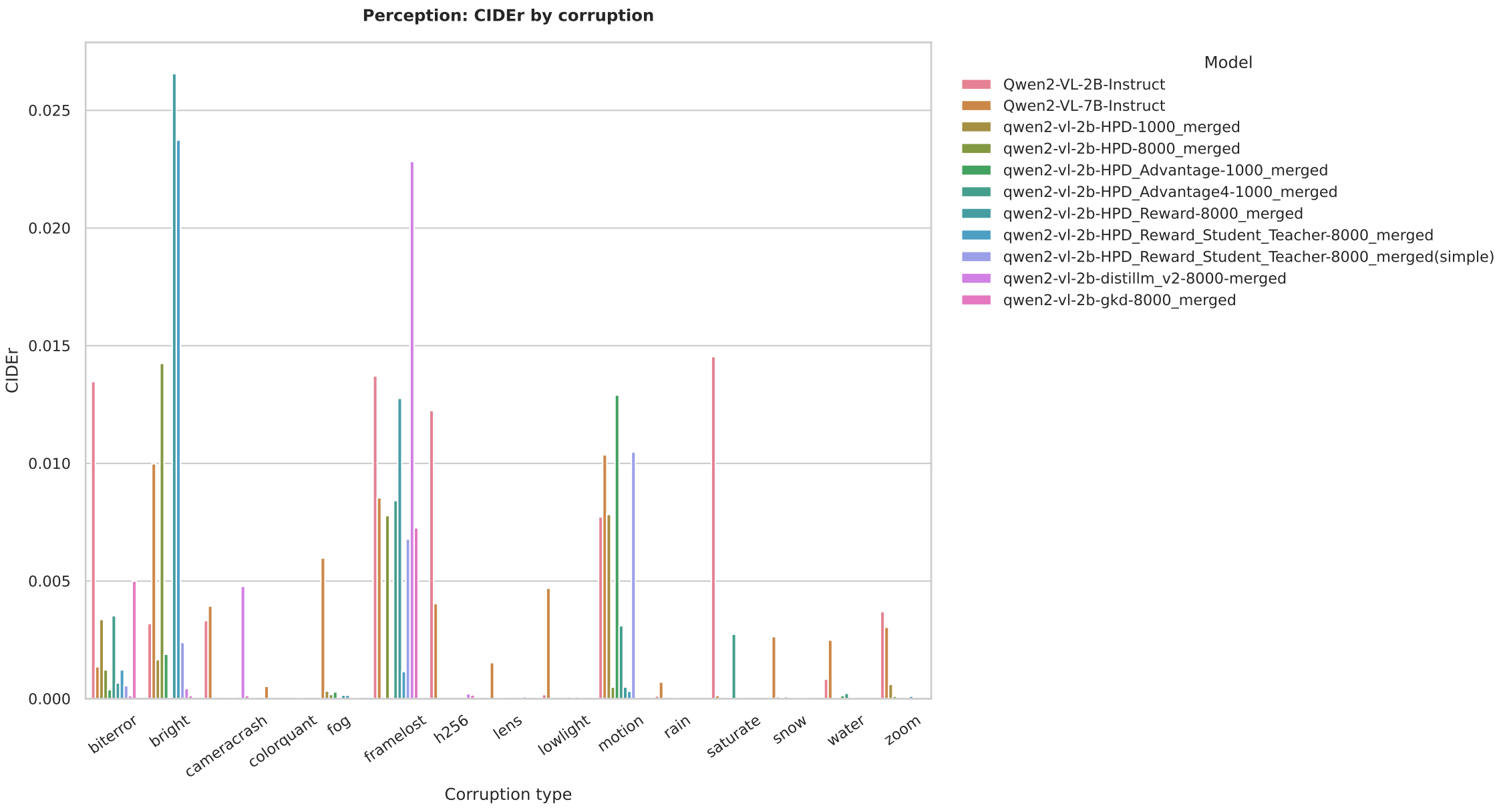
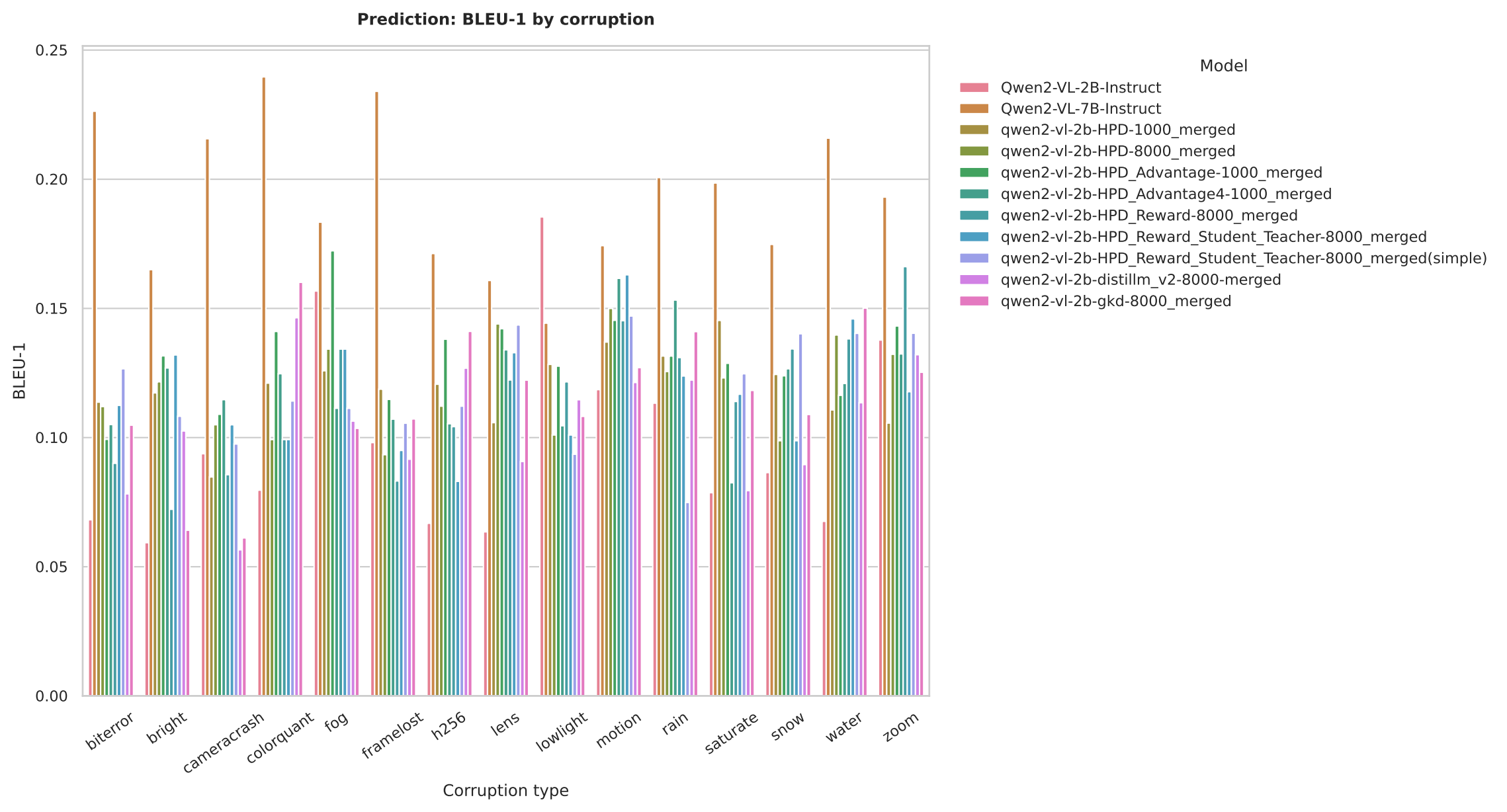

Perception: MCQ accuracy by corruption
MCQ accuracy
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

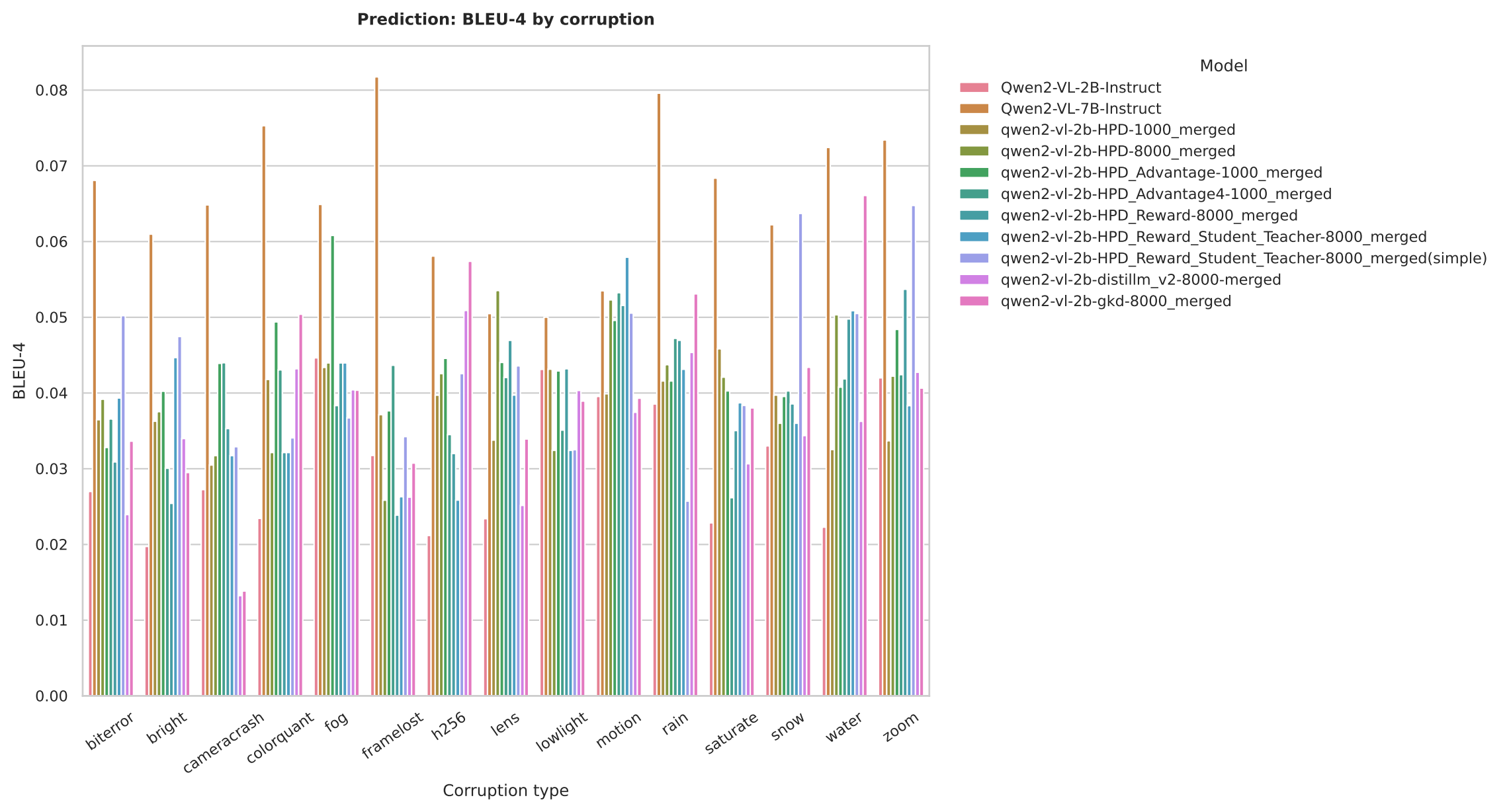

Perception: BLEU-1 by corruption
BLEU-1
0.00
0.05
0.10
0.15
0.20
0.25
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

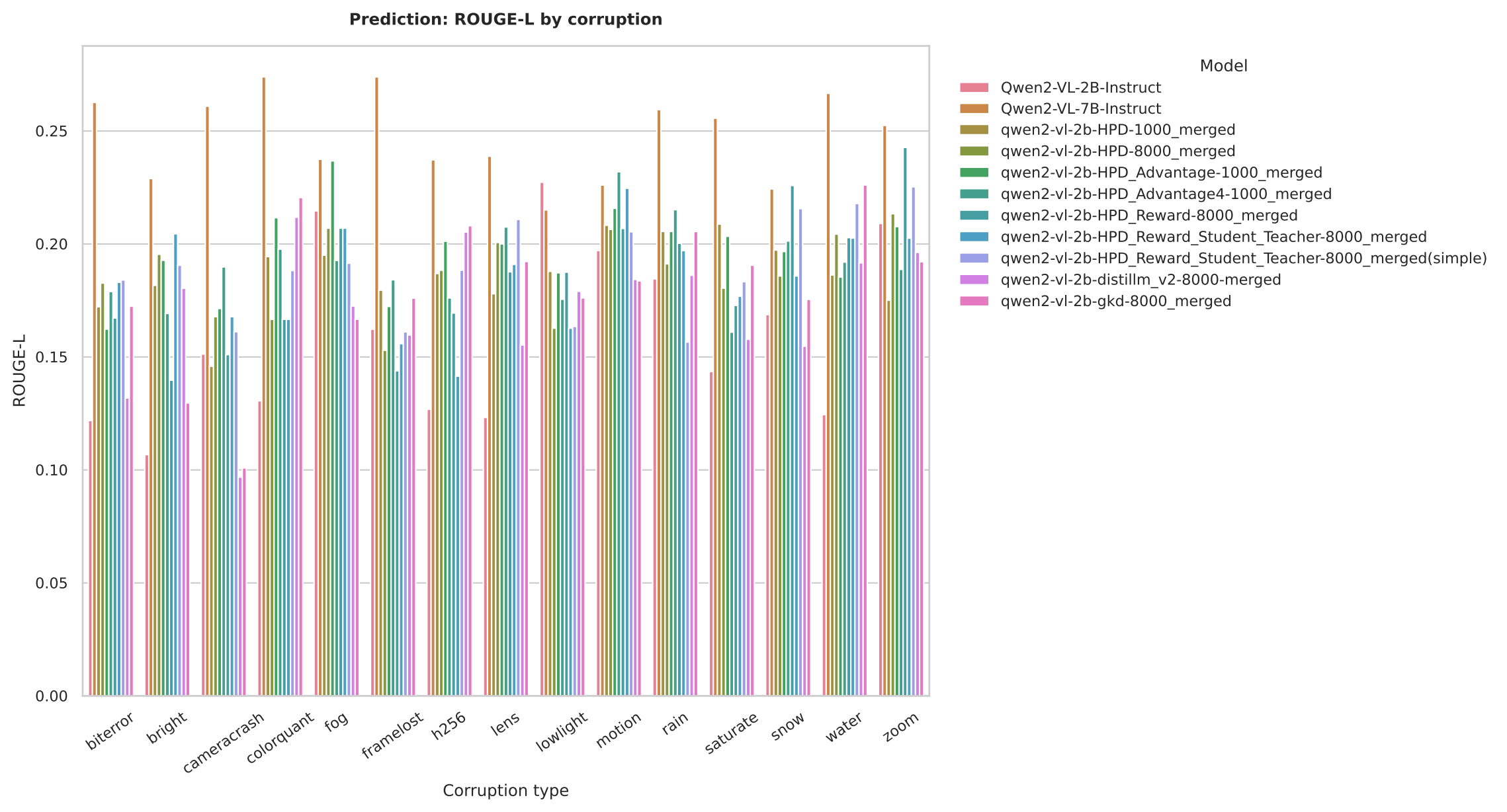

Perception: BLEU-4 by corruption
BLEU-4
0.025
0.020
0.015
0.010
0.005
0.000
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

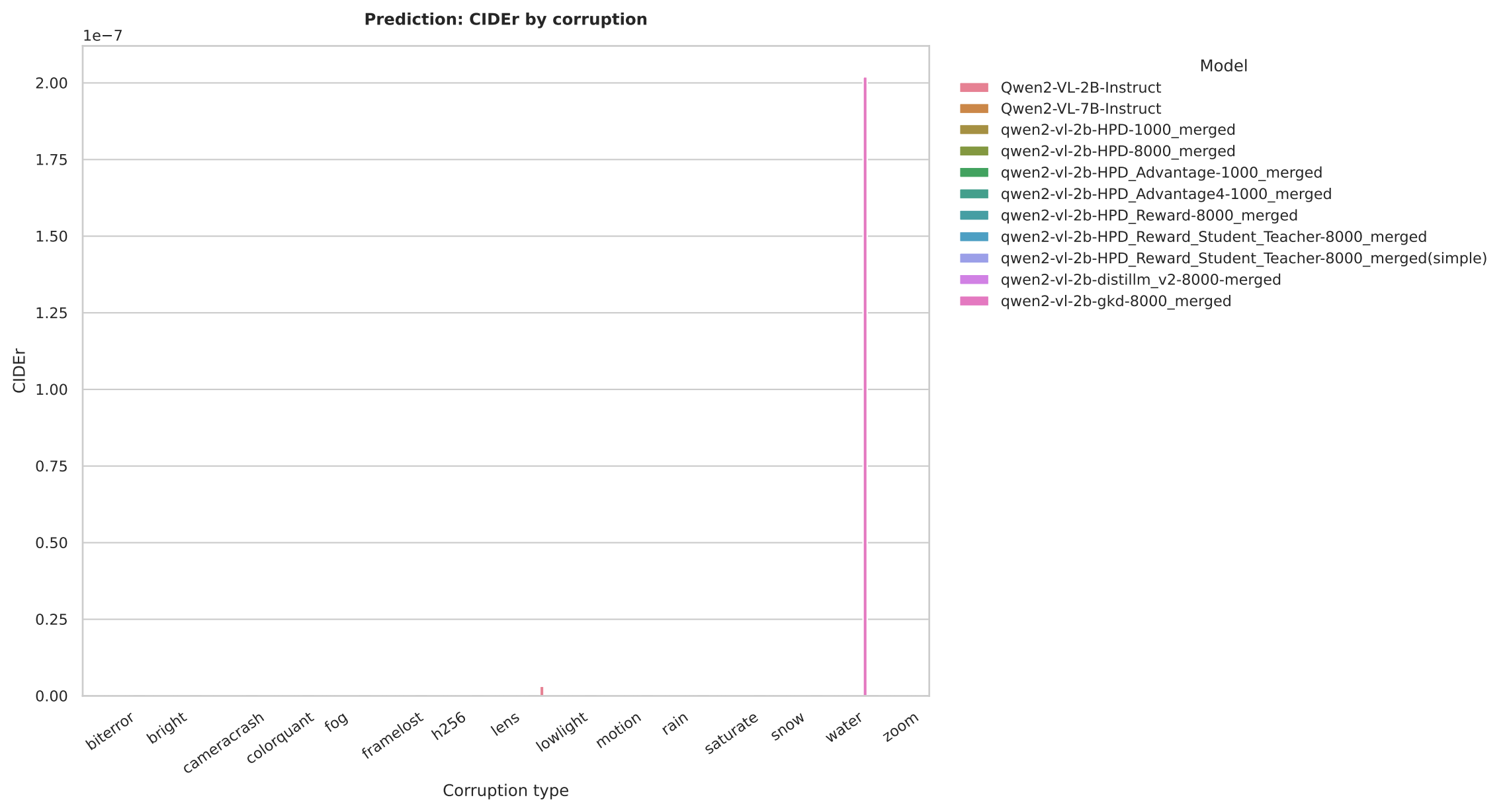

Perception: ROUGE-L by corruption
ROUGE-L
0.00
0.05
0.10
0.15
0.20
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

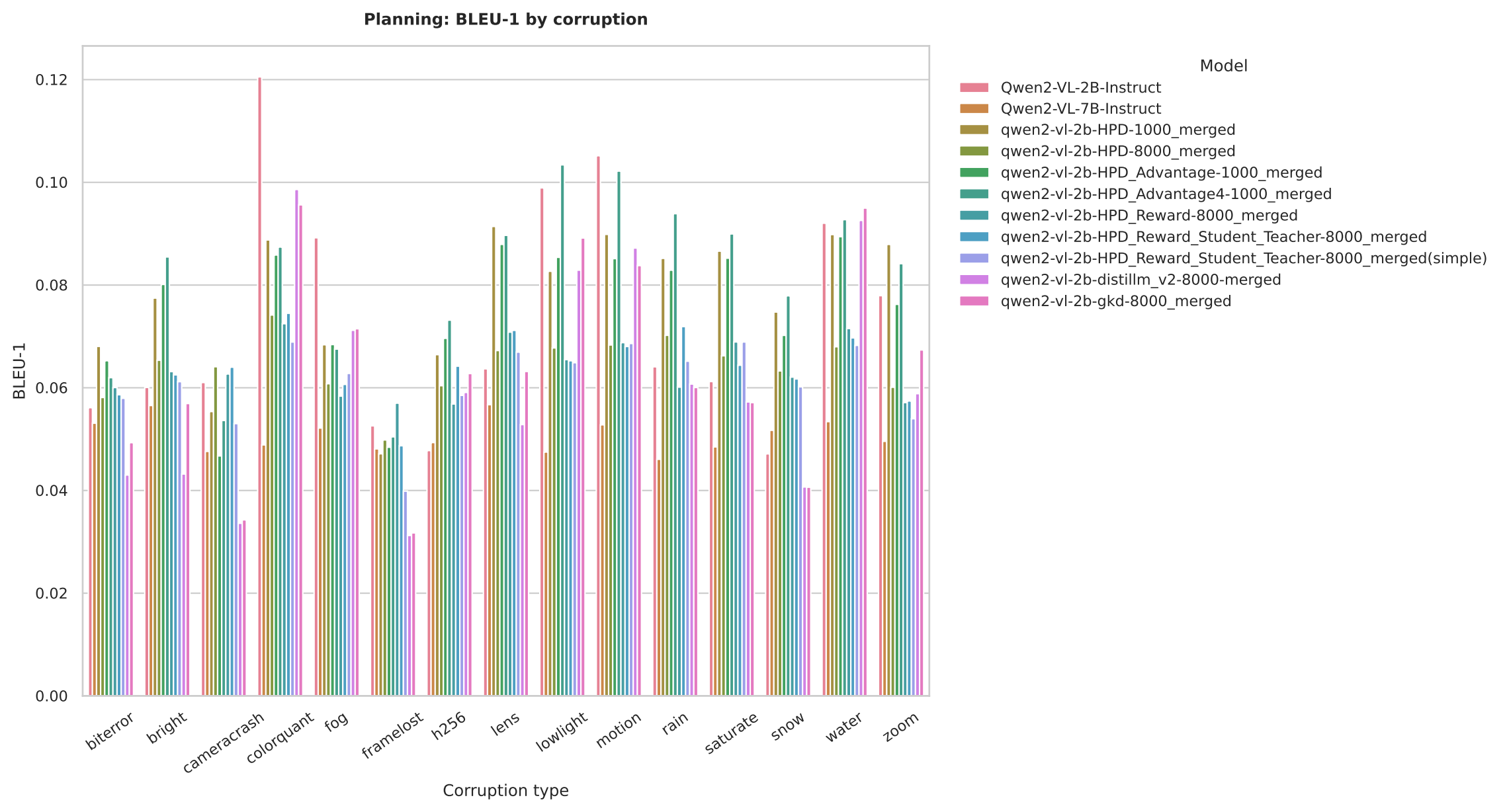

Perception: CIDEr by corruption
CIDEr
0.000
0.005
0.010
0.015
0.020
0.025
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

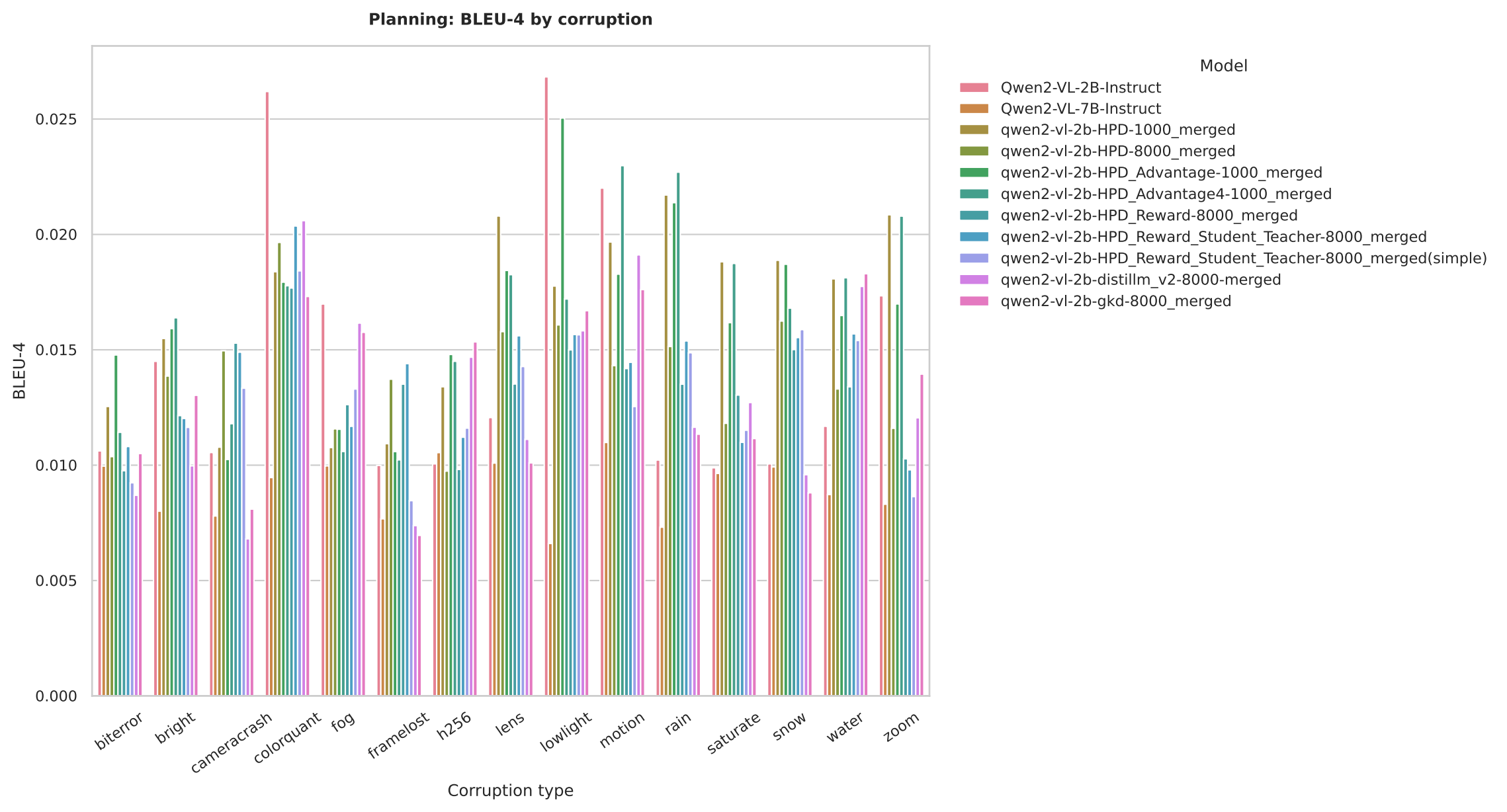

Prediction: BLEU-1 by corruption
BLEU-1
0.25
0.20
0.15
0.10
0.05
0.00
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

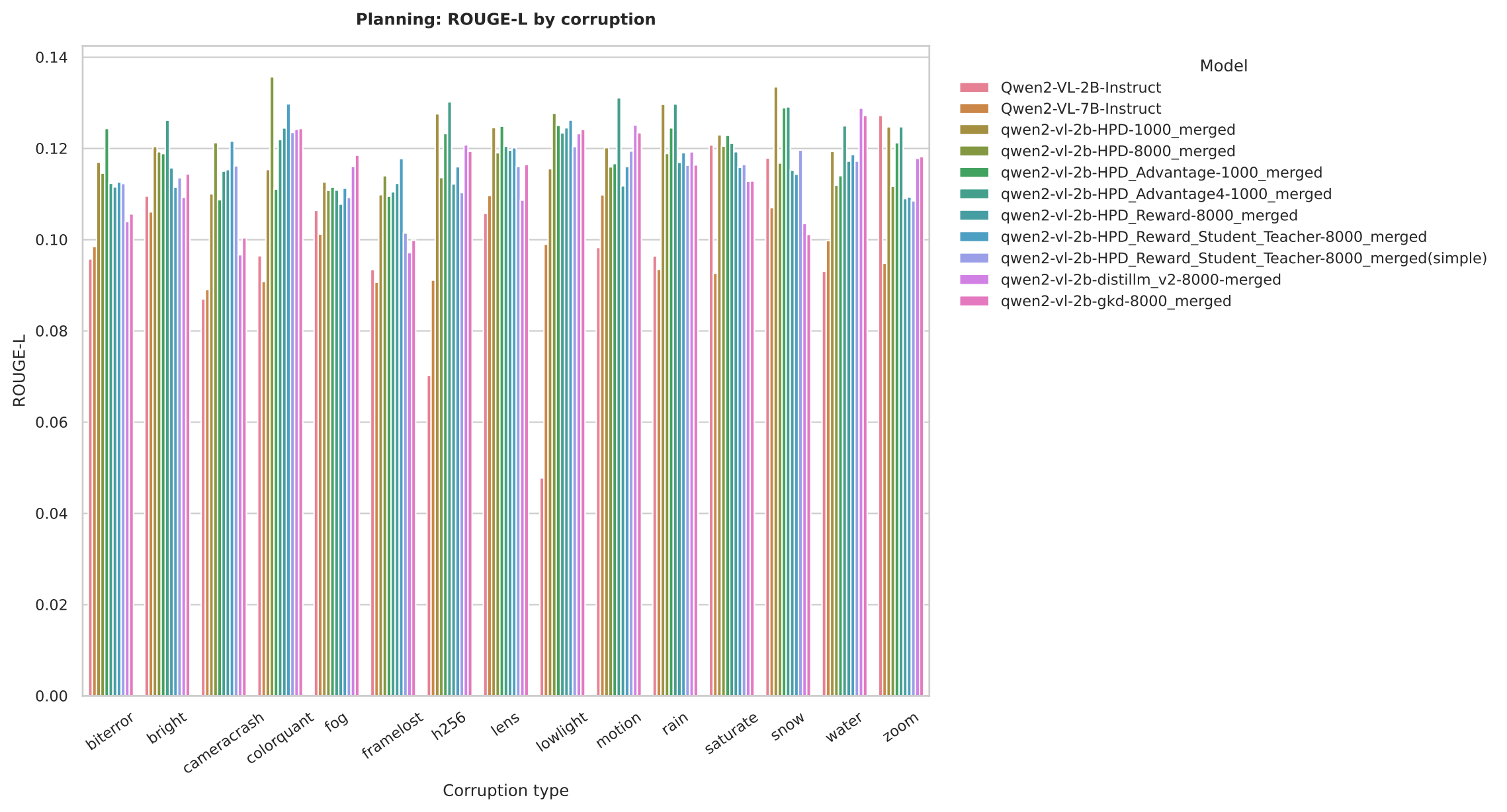

Prediction: BLEU-4 by corruption
BLEU-4
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

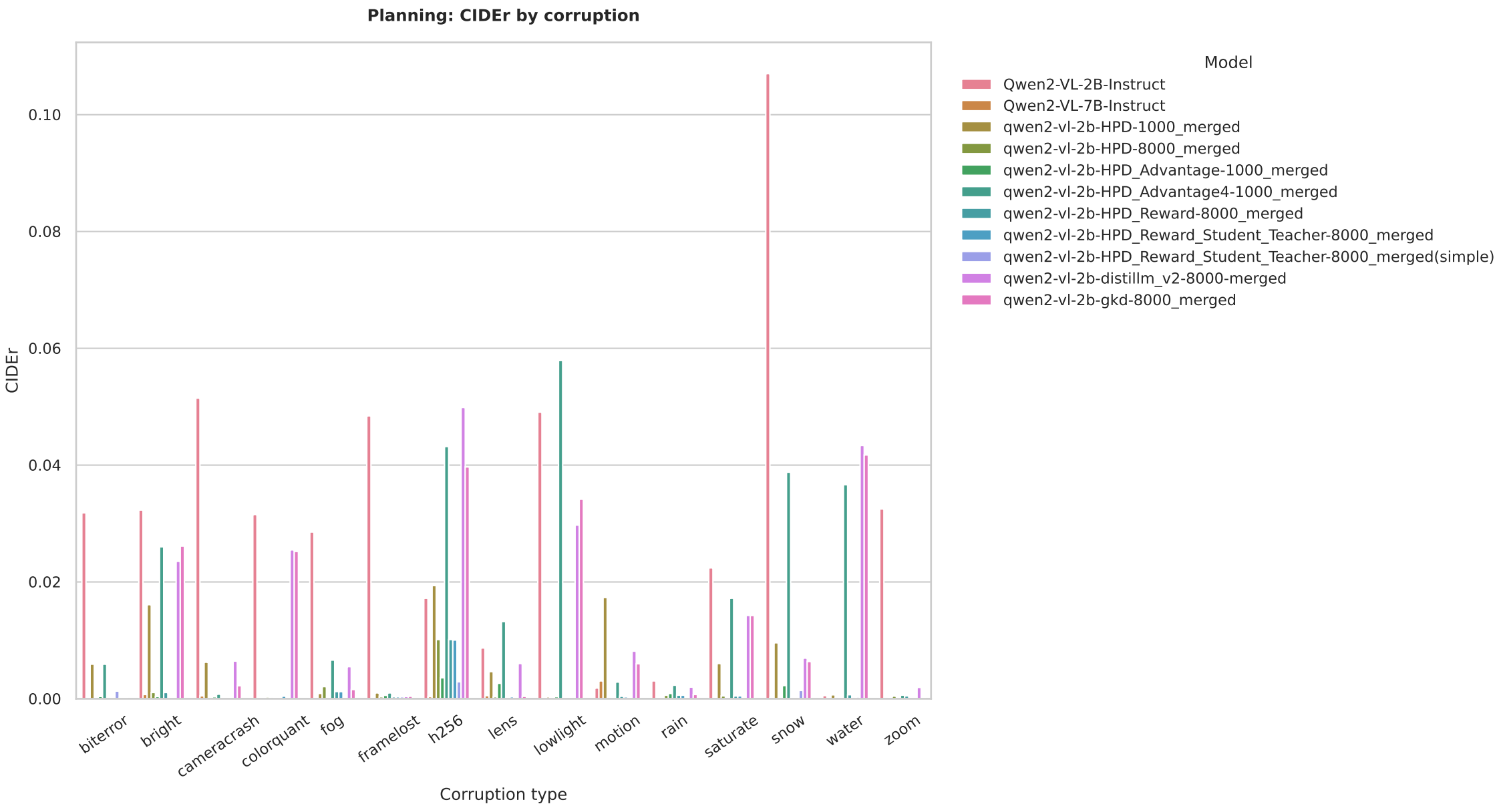

Prediction: ROUGE-L by corruption
ROUGE-L
0.00
0.05
0.10
0.15
0.20
0.25
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged

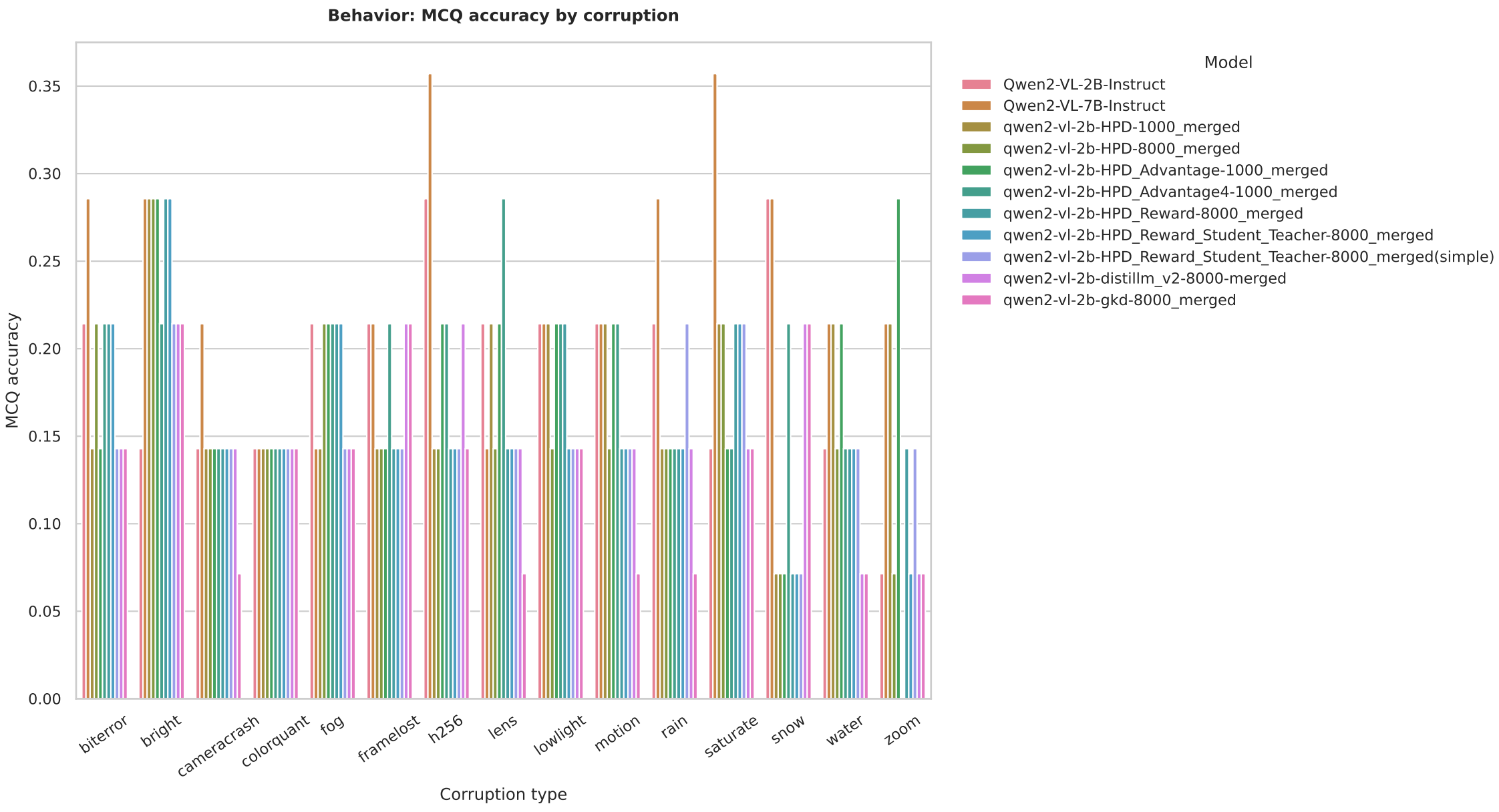

Prediction: CIDEr by corruption
1e−7
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
CIDEr
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged


Planning: BLEU-1 by corruption
BLEU-1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged


Planning: BLEU-4 by corruption
BLEU-4
0.000
0.005
0.010
0.015
0.020
0.025
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged


Planning: ROUGE-L by corruption
ROUGE-L
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged


Planning: CIDEr by corruption
CIDEr
0.00
0.02
0.04
0.06
0.08
0.10
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged


Behavior: MCQ accuracy by corruption
MCQ accuracy
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
qwen2-vl-2b-HPD-1000_merged
qwen2-vl-2b-HPD-8000_merged
qwen2-vl-2b-HPD_Advantage-1000_merged
qwen2-vl-2b-HPD_Advantage4-1000_merged
qwen2-vl-2b-HPD_Reward-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged
qwen2-vl-2b-HPD_Reward_Student_Teacher-8000_merged(simple)
qwen2-vl-2b-distillm_v2-8000-merged
qwen2-vl-2b-gkd-8000_merged